INCH-POUND

MIL-S-40059E
30 June 1989
SUPERSEDING
MIL-S-40059D
15 March 1971

MILITARY SPECIFICATION

SEPARATORS, SUSPENSION LINE, PARACHUTE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers parachute suspension line separators.

1.2 Classification. Separators shall be of the following sizes, as specified (see 6.2).

Size - Large
Size - Small

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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SPECIFICATIONS

FEDERAL

A-A-203	Paper, Kraft, Untreated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads
MIL-STD-731	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

US ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

11-1-3512 - Separator, Line, Parachute, Small
11-1-3715 - Separator, Line, Parachute, Large
11-1-3716 - Base
11-1-3717 - Separator

(Copies of drawings are available from U.S. Army Natick Research, Development and Engineering Center, ATTN: STRNC-EMSS, Natick, MA 01760-5014.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. Materials and components shall conform to the requirements specified herein and as specified on drawings 11-1-3512, 11-1-3715, 11-1-3716 and 11-1-3717. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3 Workmanship. The end item shall conform to the quality of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing operation to determine whether construction details are in accordance with specified requirements. Whenever nonconformance is noted, corrections shall be made to the items affected and lot in-process. Items which cannot be corrected shall be removed from production.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of separators of one size. The lot size shall be one separator. The inspection level shall be II (see 6.4).

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Construction and workmanship	Any component missing, malformed damaged, or not assembled as specified	101	
	Sharp edges or burrs	102	
	Rounded edges less than .015R		201
	Hardware not specified type or size improperly finished or corroded	103	
	Loose screws, nuts or hinges		202
	Hardware not installed flush or aligned	104	

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4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of separators of one size. The sample unit shall be one separator. The inspection level shall be S-3 (see 6.4).

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 (see 6.4).

Examine	Defect
Marking (exterior interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling. Bulged or distorted container.
Content	Number per container is more or less than required.

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 (see 6.4).

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation.

5.1.1.1 Small size. Ten small separators shall be nested and then wrapped with 30-pound minimum basis weight kraft paper conforming to A-A-203. The paper wrap shall be secured with tape. Forty separators (four bundles) shall be packed in an intermediate fiberboard container conforming to style RSC, grade 200 of PPP-B-636. The dimension of the box shall be approximately 10 inches in length, 9 inches in width and 10 inches in depth. The box shall be closed in accordance with method I as specified in the appendix of PPP-B–636.

5.1.1.2 Large size. Each large separator shall be folded to its smallest dimension. The folded separator shall have the base and leg components inter-leaved with a sheet of kraft paper as described in 5.1.1.1. Each separator shall be unit packed in a fiberboard container conforming to style OPF, grade 200 of PPP-B-636. The dimensions of the box shall be approximately 16 inches in length, 16 inches in width and 1 3/4 inches in depth. The box shall be closed in accordance with method I as specified in the appendix of PPP-E-636.

5.1.2 Commercial preservation. Separators shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2).

5.2.1 Level A packing. One hundred and sixty small separators or 10 large separators preserved as specified in 5.1, shall be packed in a cleated plywood container conforming to overseas type, style A or I, grade B, type 2 load of PPP-B-601. Inside dimensions of the box shall be approximately 20 1/2 inches in length, 18 1/2 inches in width and 10 1/4 inches in-depth for the small separators; and 32 3/4 inches in lengths, 16 l/2 inches in width and 10 1/4 inches in depth for the large separators. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-601.

5.2.2 Level B packing. One hundred and sixty small separators or 10 large separators preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V3c, V3s or V4s of PPP-B-636. Inside dimensions of the box shall be approximately 20 1/2 inches in length, 18 1/2 inches in width and 10 1/4 inches in depth for the small separators; and 32 3/4 inches in length, 16 1/2 inches in width and 10 1/4 inches in depth for the large separators. Each box shall be closed in accordance with method III as specified in the appendix of PPP-B-636.

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5.2.3 Commercial packing. Separators, preserved as specified in 5.1, shall be packed in accordance with ASTM D-3951.

5.3 Palletization. When specified (see 6.2), separators, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 6 for small separators, and number 94 for large separators in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, interior packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The separators are used to hold parachute suspension lines in place while the canopy in being folded. The small size separator is used to pack personnel and extraction parachutes; the large size separator is used to pack cargo parachutes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.1.1).
g. When palletization is required (see 5.3).
h. Acceptance criteria required (see 6.4).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspecsion, and approval of the first article.

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6.4 Acceptance criteria. The acceptance criteria below is recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in accordance with MIL-STD-105.

6.4.1 For end item visual examination. An acceptable quality level (AQL), expressed in terms of defects per hundred units, of 4.0 for major defects, 10 for minor defects, and 15 for total (major and minor combined) defects is recommended.

6.4.2 For end item dimensional examination. An AQL, expressed in terms of defects per hundred units, of 4.0 is recommended.

6.4.3 For packaging examination. An AQL, expressed in terms of defects per hundred units, of 2.5 is recommended.

6.4.4 For palletization examination. An AQL, expressed in terms of defects per hundred units, of 6.5 is recommended.

6.5 Subject term (key word) listing.

Packing aid

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - AS
Air Force - 99

Review Activities:
Army - AV
Air Force - 82

User Activity:
Navy - MC

Preparing activity:
Army - GL

(Project 1670-0770)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-40059E	SEPARATORS, SUSPENSION LINE, PARACHUTE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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